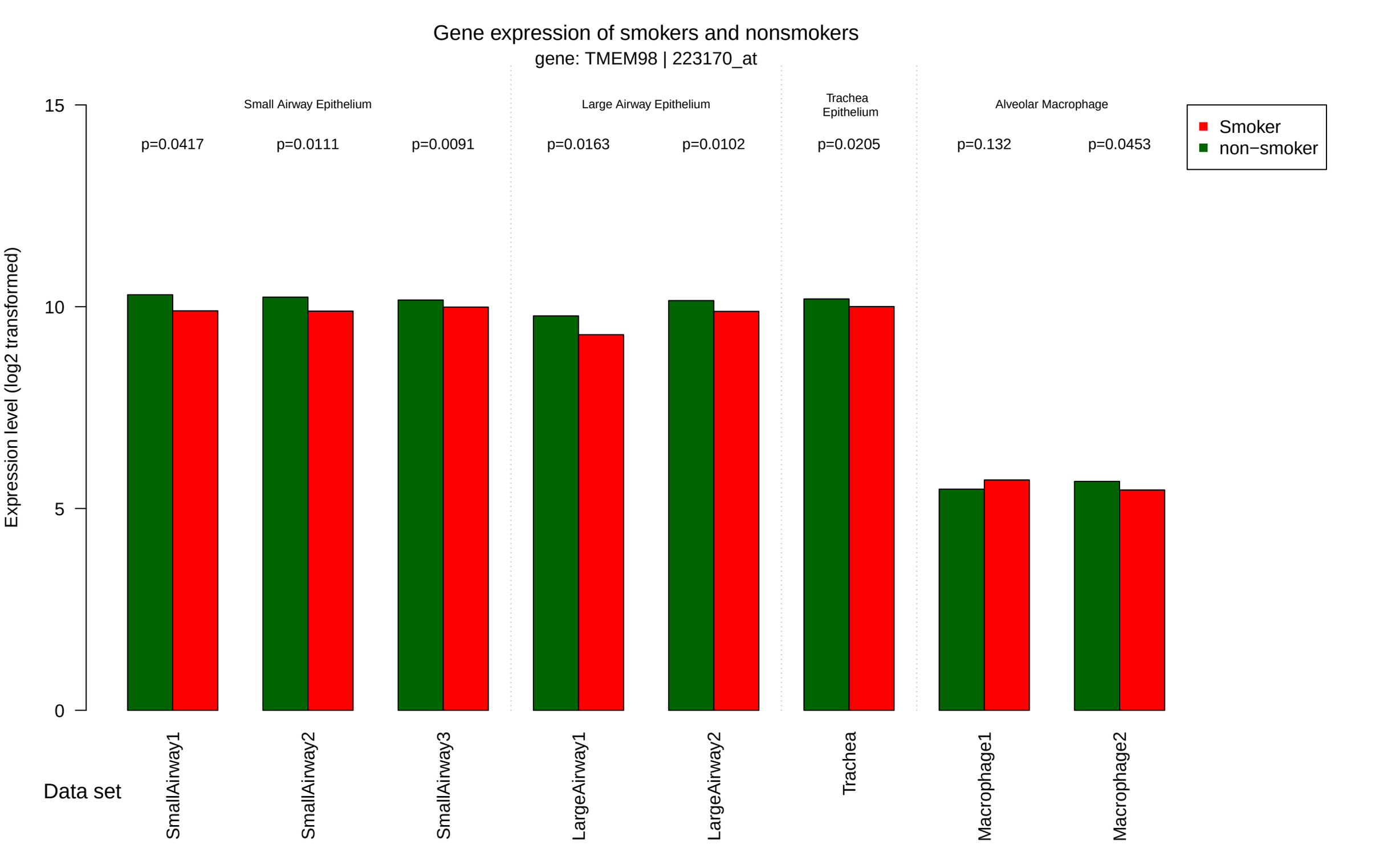
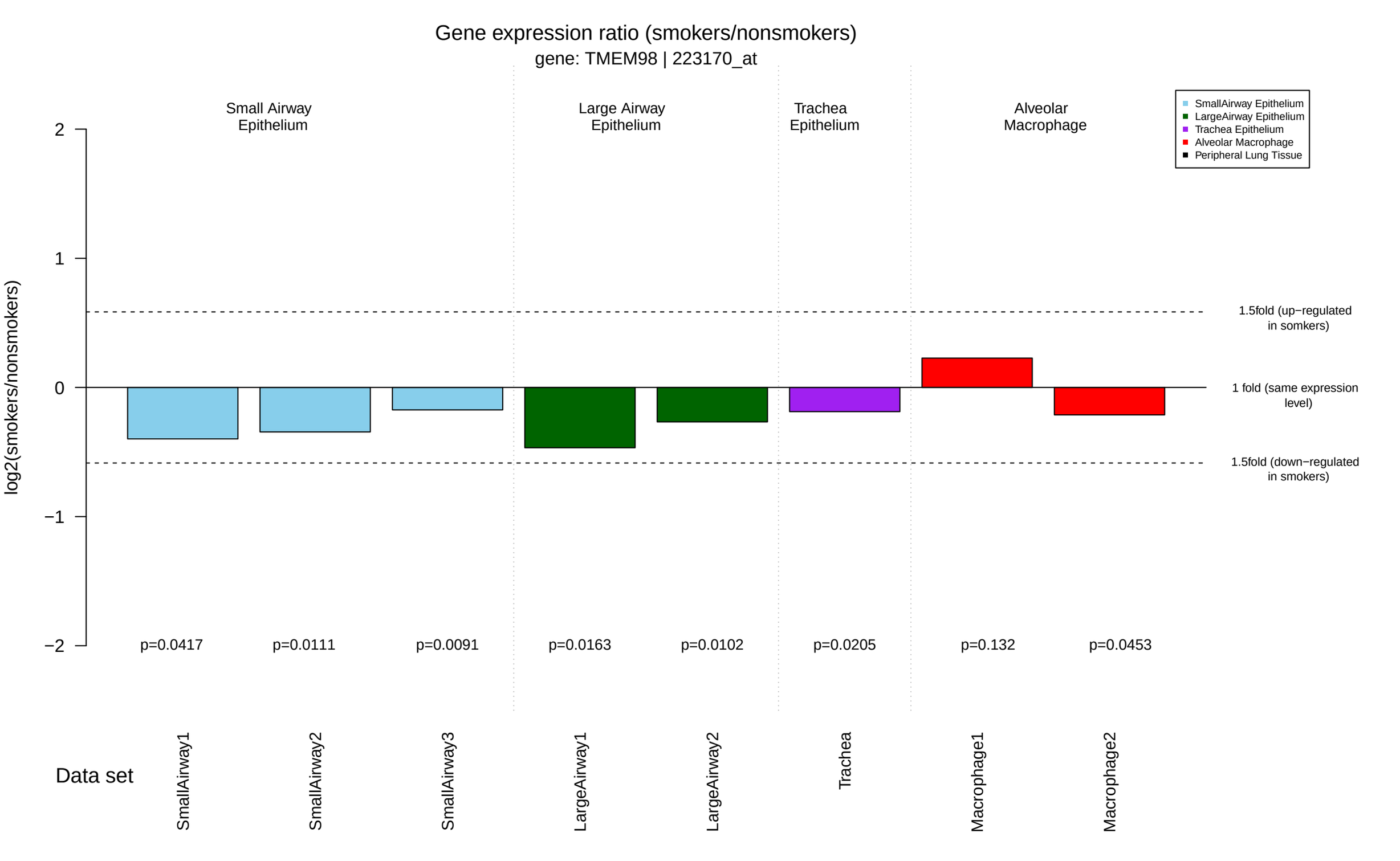
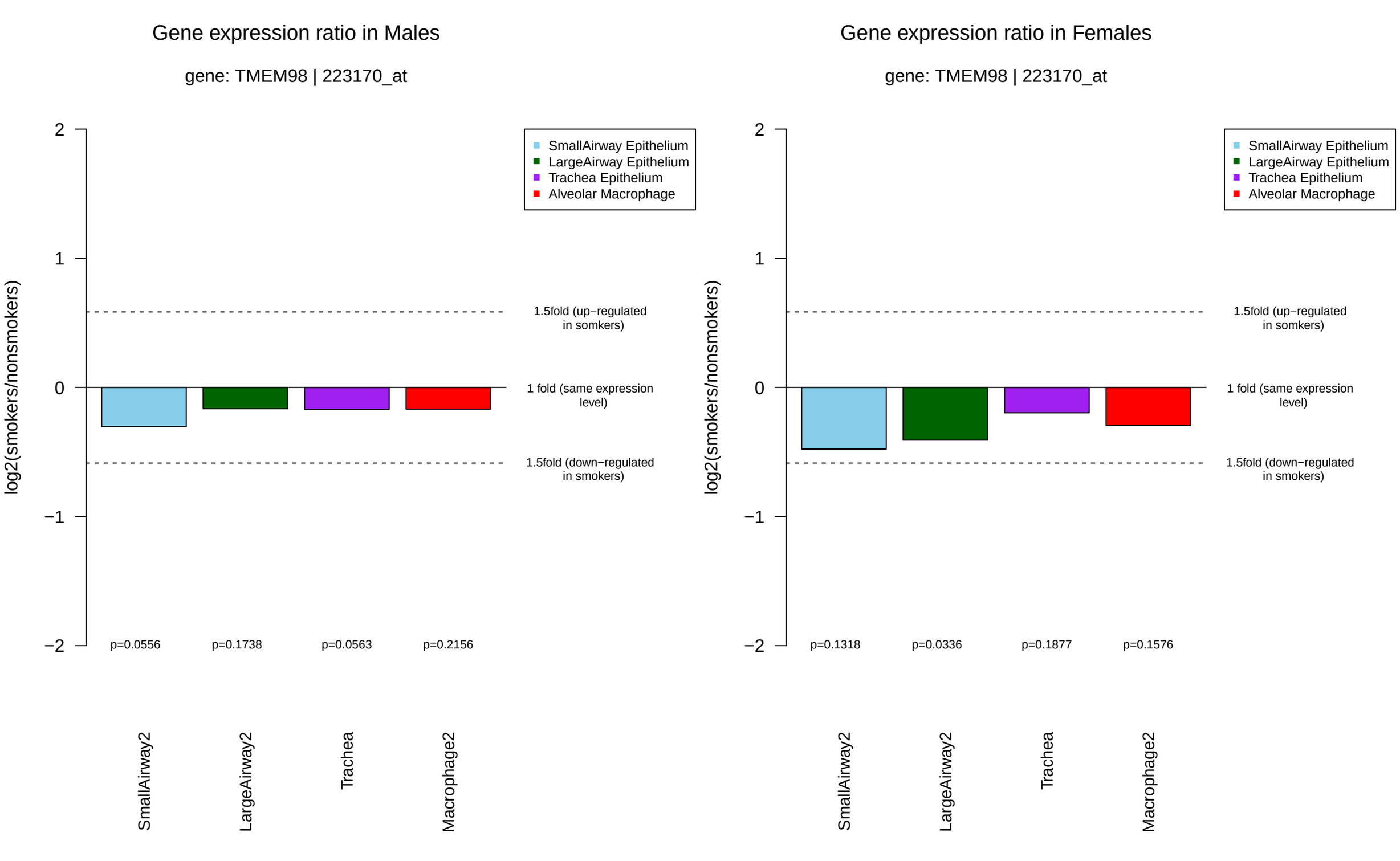
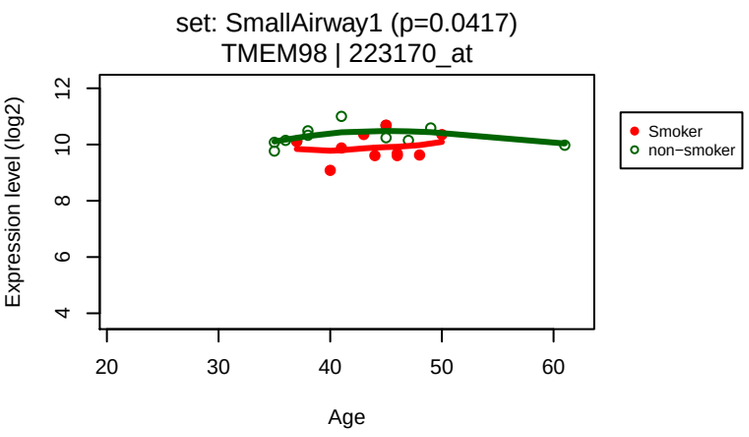
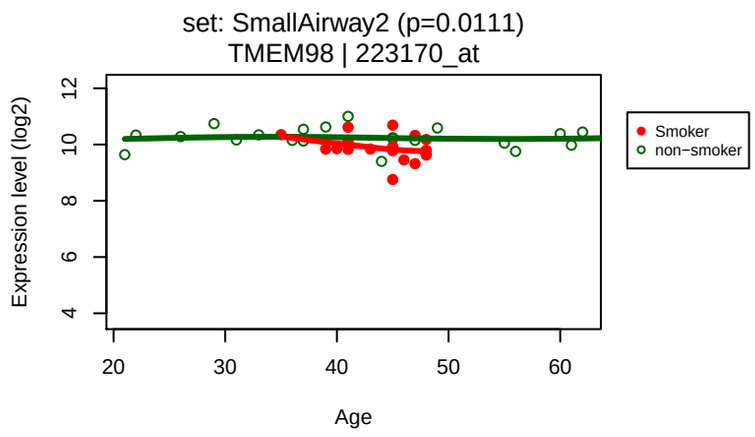
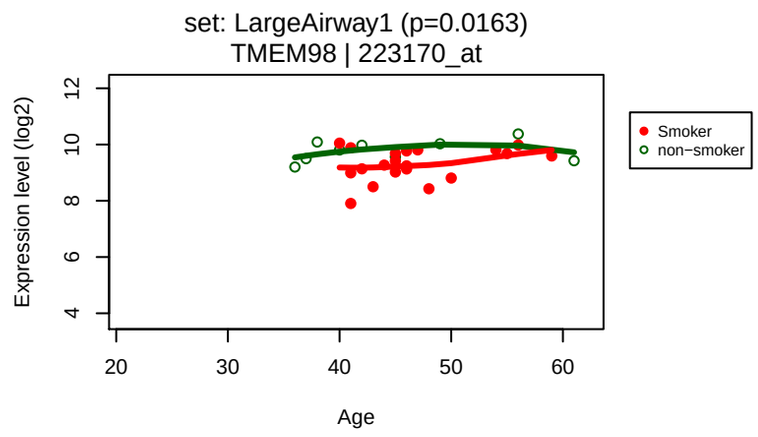

Gene expression of smokers and nonsmokers
gene: TMEM98 | 223170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0417
p=0.0111
p=0.0091
p=0.0163
p=0.0102
p=0.0205
p=0.132
p=0.0453
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

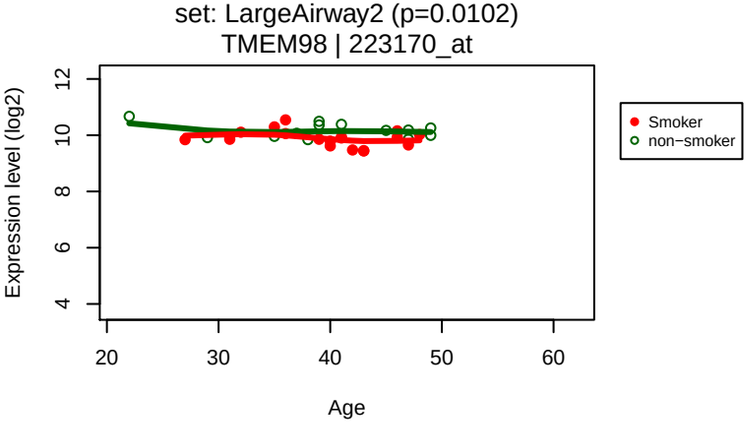

Gene expression ratio (smokers/nonsmokers)
gene: TMEM98 | 223170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.0111
p=0.0091
p=0.0163
p=0.0102
p=0.0205
p=0.132
p=0.0453
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

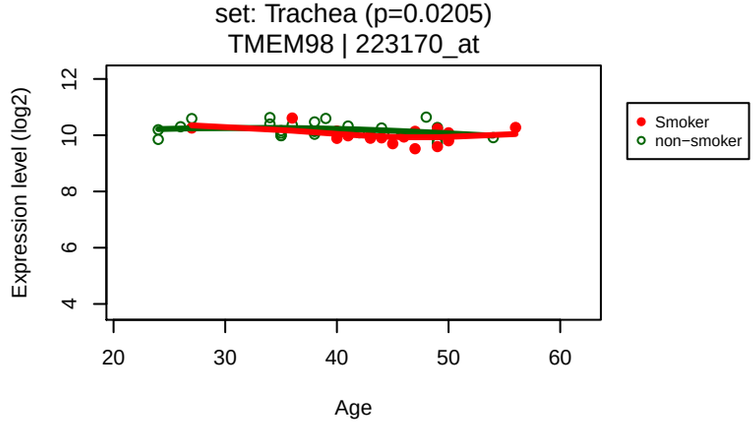

Gene expression ratio in Males
gene: TMEM98 | 223170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0556
p=0.1738
p=0.0563
p=0.2156
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM98 | 223170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1318
p=0.0336
p=0.1877
p=0.1576
SmallAirway2
LargeAirway2
Trachea
Macrophage2

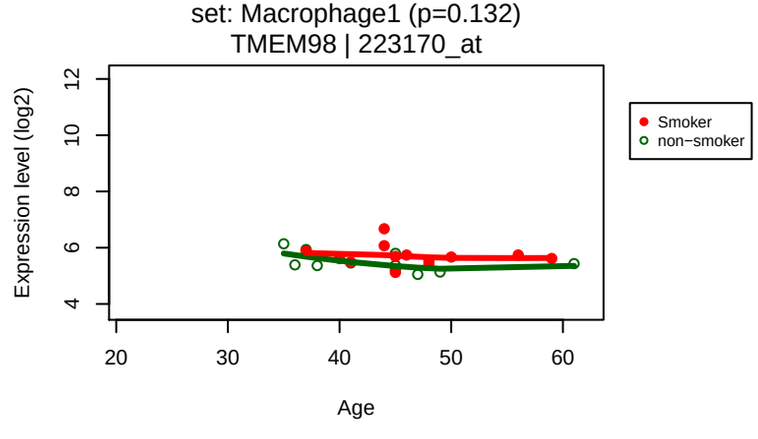

set: SmallAirway1 (p=0.0417)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker

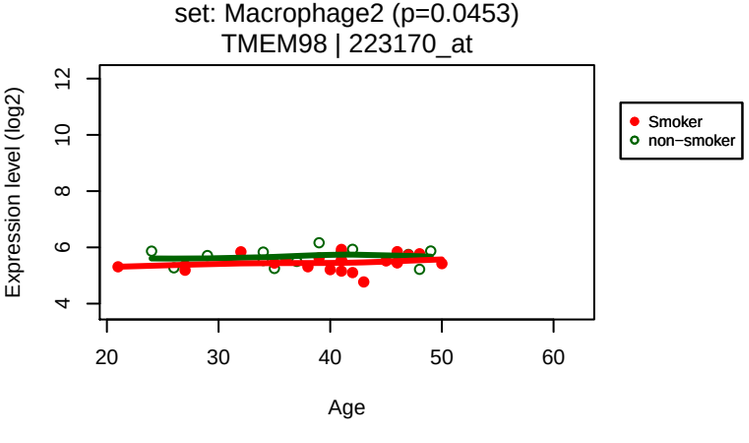

set: SmallAirway2 (p=0.0111)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker

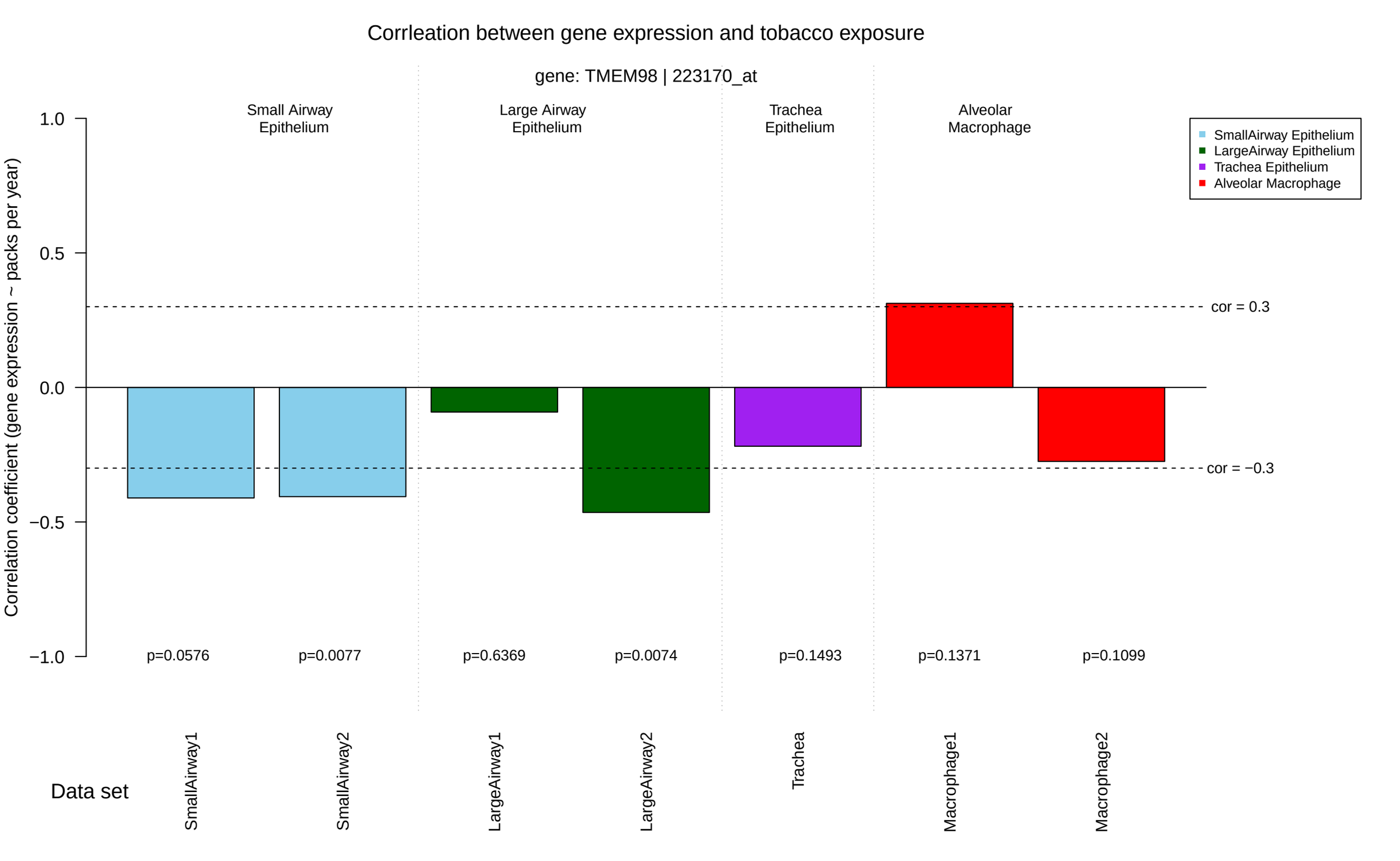

set: LargeAirway1 (p=0.0163)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0102)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0205)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.132)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0453)
TMEM98 | 223170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM98 | 223170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0576
p=0.0077
p=0.6369
p=0.0074
p=0.1493
p=0.1371
p=0.1099
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2